

ACEFA

AUSTRALIA-AOTEAROA CONSORTIUM FOR EPIDEMIC FORECASTING & ANALYTICS

ACEFA Epidemic Analytics and Modelling Short Course

12–13 November 2026

Course overview

Welcome to the ACEFA Epidemic Analytics and Modelling Short Course for 2026. The Australia–Aotearoa Consortium for Epidemic Forecasting and Analytics (ACEFA) is an NHMRC-funded Centre of Research Excellence that brings together researchers, public health agencies, and partner organisations across Australia and Aotearoa New Zealand to strengthen preparedness and response to epidemic diseases through real-time data analytics, infectious disease modelling, and forecasting.

We are delighted to be working with the Australian Centre for Disease Control (CDC) to deliver this two-day short course, which provides an introduction to the principles and applications of infectious disease modelling in public health. Through a combination of lectures, practical exercises, and discussion, the course will explore how models can be used to understand disease transmission, interpret surveillance data, evaluate interventions, and support policy development. We look forward to engaging with Australian CDC colleagues and to fostering ongoing collaboration between modelling researchers and public health policymakers.

The material in this workbook was prepared by ACEFA for the Epidemic Analytics and Modelling Short Course, delivered to participants at the Australian Centre for Disease Control on 12–13 November 2026. We kindly request that the material is not reproduced or distributed.

Presenters and contributors

Associate Professor Freya Shearer

Freya Shearer is an infectious disease epidemiologist and mathematical modeller at the University of Melbourne, where she co-leads the Infectious Disease Dynamics Unit. Her research combines mathematical modelling, statistics, surveillance analytics and decision science to improve understanding of infectious disease transmission and inform public health policy. She has played a leading role in Australia's response to emerging infectious diseases, providing modelling and epidemiological advice to governments and public health agencies during the COVID-19 pandemic and Japanese encephalitis outbreak (2021–23). She completed her PhD at the University of Oxford and has established an internationally recognised research program focused on delivering analytical tools and evidence to support public health decision-making in Australia and globally. She is Director and CIA of ACEFA.

Dr Oliver Eales

Oliver Eales is a University of Melbourne McKenzie Fellow with a joint appointment across the Melbourne School of Public Health and the School of Mathematics and Statistics. He was awarded a PhD from Imperial College London in March 2023. During his PhD, he was a lead mathematical modeller for the REACT-1 study, one of only two national-scale infection prevalence studies of COVID-19 in the world. His research is in the development and application of mathematical and statistical models to inform surveillance and response to infectious diseases—including respiratory pathogens, vector-borne diseases, sexually transmitted diseases, and zoonotic diseases. He is particularly interested in research translation; as Reporting Lead for ACEFA's Winter Situational Assessment Program, he synthesises modelling outputs from teams across Australia and New Zealand into weekly policy-relevant reports that are delivered to the Australian Centre for Disease Control, state Departments of Health and the New Zealand Ministry of Health.

Dr Alexandra Hogan

Alexandra Hogan completed her PhD in mathematical infectious disease modelling at the Australian National University in 2017, which was followed by postdoctoral training and a research fellowship at Imperial College London. Since 2022, she has been a Senior Scientia Lecturer at UNSW Sydney, and co-leads the Infectious Disease Dynamics and Interventions research group. Alexandra works on developing transmission models for respiratory pathogens, with a focus on estimating the population-level health impact of new vaccines, including for RSV, influenza and SARS-CoV-2. She previously worked on modelling to support the World Health Organization recommendation on use of the RTS,S malaria vaccine, and on informing global rollout of COVID-19 vaccines. Alexandra has a particular interest in translating modelling outputs to guide policy. She is a member of the WHO Immunisation and Vaccines Implementation Research Advisory Committee, and the NCIRS Scientific Advisory Committee.

Dr Richard Cresswell

Richard Creswell is a research fellow at the Melbourne School of Public Health, where he specializes in epidemic analytics and the design of respiratory virus surveillance systems. His research involves the use of mathematical modelling, Bayesian statistics, and machine learning to learn more about biological systems and gain practical insights from challenging datasets. Amongst other topics, he takes particular interests in complex time series and infectious diseases. Prior to joining the University of Melbourne in 2024, Richard completed doctoral studies in Computer Science and worked as an Eric and Wendy Schmidt "AI in Science" Fellow at the University of Oxford. Throughout his research career he has worked closely with a multitude of government, healthcare, industry, and startup collaborators.

Professor James McCaw

James McCaw is a mathematical biologist and epidemiologist and was an awarded an Australian Research Council Laureate Fellowship in 2024. He obtained a PhD in theoretical physics in 2005, before moving into the field of infectious disease dynamics where he was instrumental in establishing one of Australia's leading research groups in infectious disease dynamics and modelling at the (now) Melbourne School of Public Health. Since 2015 he has held a teaching and research position split between the School of Mathematics and Statistics and the Melbourne School of Public Health. His research interests span from the study of within-host dynamics to epidemiology and public health. During the COVID-19 pandemic, he sat on Australia's highest level advisory committee and co-led two national consortia providing model-based advice to government. He is a current invited expert member of the Communicable Diseases Network of Australia.

Dr Wang Jin

Wang Jin is a Lecturer in the School of Population Health at the University of New South Wales and an infectious disease modeller. He previously worked as a research fellow at Queensland University of Technology and the Kirby Institute. With a background in applied mathematics, his research combines mathematical modelling, statistical methods and evidence synthesis to investigate pathogen dynamics, disease transmission, immune protection and vaccine effectiveness. His research interests include infectious disease modelling and the meta-analysis of aggregate evidence in public health, with the aim of translating quantitative evidence into improved disease control and health policy. His research has spanned collective cell behaviour in cancer biology, HIV infection, RSV transmission and SARS-CoV-2 vaccine effectiveness.

Professor James Wood

Professor James Wood is Head of the School of Population Health at UNSW Sydney and an infectious disease researcher whose work draws on a PhD in applied mathematics to understand disease epidemiology and predict the population-level effects of health interventions, particularly vaccines. His research spans pandemic preparedness, immunisation programs, respiratory virus epidemiology, sexually transmitted infections, surveillance systems, and public health interventions. He played significant roles in Australia's COVID-19 response, including service on the Australian Technical Advisory Group on Immunisation and modelling work for NSW, Australia, and the WHO Western Pacific Region. He also contributed to Australia's original pandemic planning and vaccine rollout modelling during the 2009 H1N1 pandemic. His current research program focuses on infectious disease dynamics and interventions, novel surveillance tools, and pandemic preparedness.

Course outline

Mathematical models are critical tools for understanding the spread of infectious diseases through populations, and the potential impact of public health interventions. When combined with statistical analysis, modelling is a useful tool for interpreting and synthesising noisy, incomplete, and unrepresentative surveillance data. In this course you will learn about the foundations of epidemiological modelling and data analytics in the context of epidemic respiratory viruses and vaccine preventable diseases.

Day 1: Introduction to Modelling and Epidemic Theory

Lecture 1: Introduction to modelling: Infectious diseases and mathematical models
This lecture introduces key concepts and processes in infectious disease modelling for public health. We will include real-world examples of uses of modelling in public health research and policy development. We will also discuss key considerations in model development, including model complexity, parameterisation, uncertainty and sensitivity analysis.

Lecture 2: Epidemic theory part 1: The SIR model and typical epidemic dynamics
This lecture introduces key concepts in infectious disease modelling. We will introduce and analyse the behaviour of the SIR model, the most well known compartmental model.

Practical 1: Exploring SIR model dynamics
In this interactive practical session, we will explore the outputs of an SIR model that is implemented as a set of difference equations in Microsoft Excel. Using the model, you will be guided through a series of questions that allow you to investigate the properties of the model, and how the outputs changes with different parameter assumptions. You will also compare deterministic and stochastic versions of the model, and see in practice how the reproduction number is related to key epidemiological measurements such as the final epidemic size and timing of the epidemic peak.

Lecture 3: Epidemic theory part 2: Epidemiological and biological parameters and their interactions with transmission and interventions
This lecture introduces key epidemiological quantities that underpin infectious disease transmission dynamics, including the generation interval and serial interval. We will cover how these quantities relate to epidemic growth and dynamics over time, and how individual-level heterogeneity in contact patterns and other characteristics influences how diseases spread. The lecture will also introduce how disease models are used to predict the impact of interventions, focussing on vaccination, through practical examples drawn from COVID-19 and RSV immunisation policy.

Practical 2: Implementing SIR Models in R
In this practical, we will again examine the deterministic and stochastic SIR models, but instead implement these in R as differential equations. We solve the set of differential equations numerically to produce trajectories of infections (and other modelled states) over time. This allows us to see how the implementation of models looks different when translated from a spreadsheet into R code, and understand how analysis (particularly if tracking many scenarios, or trajectories over longer time periods) can become somewhat easier in R.

Day 2: Introduction to Analysis of Epidemic Time-series Data

Lecture 4: Introduction to time-series analysis: Quantifying temporal trends in epidemic data
This lecture will introduce two mathematical formulations for quantifying how transmission dynamics change over time: the epidemic growth rate, and the effective reproduction number. We will cover how these quantities are related to the time-series of infection incidence.

Lecture 5: Biases in the disease observation process
This lecture will describe common biases in epidemic time-series that arise from the observation processes of surveillance systems. We will provide an overview of the methods that exist for accounting for these data problems, emphasising the importance of maintaining data integrity so that such methods can be used.

Practical 3: Real-time epidemic analysis: Estimating growth rates and reproduction numbers using a statistical model
Practical 3 will run in multiple parts over the second day of the course. The practical will provide hands-on experience in analysing epidemic time-series, quantifying their transmission dynamics, and visualising and interpreting the results. The practical will use the R package EpiNow2 — this should be installed before the second day of the course.

Lecture 6: Quantifying trends in multiple pathogens, strains, or variants from composite time-series data
This lecture will introduce further biases in epidemic time series from routinely collected surveillance data. We will focus on how trends in many routine surveillance indicators can be biased by multiple circulating pathogens/subtypes, describing methods that exist for quantifying the distinct trends of all component pathogens (when suitable data is available). Code containing an introduction to implementing these methods in R will be provided but will not be covered in the two-day course.

Schedule

Time	Day 1 – Introduction to Modelling and Epidemic Theory		
9:30–9:45	Welcome, Housekeeping, Acknowledgement of Country, Introductions		
9:45–10:30	Lecture 1	Introduction to Modelling	Freya Shearer
10:30–10:50	Morning tea		
10:50–11:50	Lecture 2	Epidemic Theory Part 1	Oliver Eales
11:50–1:00	Practical 1	SIR Model Spreadsheet Practical	Freya Shearer
13:00–14:00	Lunch		
14:00–15:00	Lecture 3	Epidemic Theory Part 2	Alexandra Hogan
15:00–15:20	Afternoon tea		
15:20–16:20	Practical 2	Implementing SIR Models in R	Alexandra Hogan
16:20–17:00	Optional	Optional: EpiNow2 installation check and R Refresher session	Oliver Eales, Alexandra Hogan, Freya Shearer

Time	Day 2 – Analysis of epidemic time-series data		
9:00–10:00	Lecture 4	Quantifying temporal trends	Oliver Eales
10:00-10:30	Practical 3: Part 1	Quantifying trends in epidemic time-series: Part 1	Oliver Eales
10:30–10:50	Morning tea		
10:50–11:30	Practical 3: Part 2	Quantifying trends in epidemic time-series: Part 2	Oliver Eales
11:30–12:30	Lecture 5	Biases in the observation process	Richard Creswell
12:30–13:00	Practical 3: Part 3	Quantifying trends in epidemic time-series: Part 3	Oliver Eales
13:00–14:00	Lunch		
14:00–15:00	Lecture 6	Quantifying trends in multiple pathogens, strains or variants from composite time-series data	Oliver Eales
15:00–15:20	Afternoon tea		
15:20–16:20	Practical 3: Part 4	Quantifying trends in epidemic time-series: Part 4	Oliver Eales
16:20–16:30	Close	Closing comments Feedback survey	

Instructions for installing R and associated packages

Prior to the short course participants should ensure that RStudio has been installed on their laptops and that all required R packages have been installed.

Packages can be installed with the R command:

```
install.packages('PACKAGE_NAME')
```

Where package name should be provided in the quotation marks.

To check whether the package has been successfully installed you can run the R command:

```
library('PACKAGE NAME')
```

If this does not throw any errors the package has been successfully installed.

The packages required for the practicals in this course are:

- 'ggplot2'
- 'patchwork'
- 'EpiNow2'
- 'tidyverse'
- 'deSolve'

Glossary of key terms

Term	Explanation
Basic reproduction number, R_0	The expected number of secondary infections generated by one infected individual in a completely susceptible population.
Effective reproduction number, R_e	The expected number of secondary infections generated by an infected individual in a population which may not be entirely susceptible and/or in the presence of interventions.
The force of infection, commonly denoted λ	The rate at which susceptible individuals acquire an infectious disease.
The epidemic growth rate, r(t)	The rate at which infections increase over time, assuming growth is exponential over short time-scales.
Prevalence of infection	The fraction of the population infected over a specified time period.
Incidence of infection	The rate at which new infections occur over a specified time period.
Attack rate	The proportion of a population that becomes infected during a specified time period.
Compartmental models	Compartmental models are a mathematical framework for simulating how populations move between different states or “compartments”. For modelling infectious diseases the population is divided into compartments based on their epidemiological status.
SIR model	One of the simplest (and most famous) compartmental models. The population is divided into three epidemiological states: Susceptible (S), Infected and infectious (I), and Recovered/removed (R).
Deterministic models	In deterministic models the flow between compartments is set by rate parameters.There is a predetermined relationship between model inputs and model outputs.
Stochastic models	In stochastic models transitions of individuals between compartments occurs with probabilities (determined using the rate parameters) and so includes inherent randomness in the epidemic dynamics. Each simulation of a model will return a different output for the same input parameters.
Incubation period	The time between exposure and onset of symptoms (within a single individual).
Latent period	The time between exposure and becoming infectious (within a single individual).
Infectious period	The duration that an individual is infectious.
Generation interval	The time between acquisition of infections in a primary and a secondary case (i.e. time between infection acquisitions in an infector-infectee pair).
Serial interval	Time between symptom onsets in a primary and secondary case (i.e. time between symptom onsets in an infector-infectee pair).

Delay distributions	There are often delays between two outcomes of interest (e.g. infection and hospitalisation). A delay distribution will describe the probability that this time delay is of a specific length.
Generation interval distribution	The probability density function of the generation intervals across all transmission events.
Herd immunity threshold	The herd immunity threshold is the proportion of the population that must be protected (or immune) for transmission to decline (or for each infection to cause less than one new infection).

Key equations

1. The SIR model

The SIR model without demography (i.e. no births and deaths) is governed by the system of ordinary differential equations:

$$\frac{ds}{dt} = -\beta si,$$
$$\frac{di}{dt} = \beta si - \gamma i,$$
$$\frac{dr}{dt} = \gamma i,$$

with

$$s + i + r = 1.$$

Here s is the proportion of the population susceptible, i is the proportion of the population infected and infectious, and r is the proportion of the population recovered/removed. The constants β and γ represent the transmissibility rate and the recovery rate respectively.

2. Key epidemiological quantities

For the SIR model the basic reproduction number is given by the expression

$$R_0 = \frac{\beta}{\gamma}$$

(i.e. it is the daily transmission rate, multiplied by the days an individual is infectious).

The effective reproduction number is given by:

$$R_e(t) = \frac{\beta}{\gamma} \times s(t)$$

The generation interval is the delay between acquisition of infections in a primary and secondary case (i.e. between an infector-infectee pair). The generation interval is related to the reproduction number by the following equation:

$$R_0 = 1 + rT_C$$

Where r is the growth rate in cases per unit time, and T_C is the mean generation interval.

The herd immunity threshold is the proportion of the population that must be protected (or immune) for transmission to decline (or for each infection to cause less than one new infection). It is defined as

$$p_c = 1 - 1/R_0$$

3. Time-series analysis

3.a Reproduction number formulation

The infection incidence at time, $I(t)$, can be written in terms of the infection incidence at earlier times, the generation interval as a function of time since infection, $g(\tau)$, and the effective reproduction number, $R_e(t)$:

$$I(t) = R_e(t) \sum_{\tau=0}^{\infty} I(t-\tau) g(\tau)$$

This can be rearranged to give the effective reproduction number:

$$R_e(t) = \frac{I(t)}{\sum_{\tau=0}^{\infty} I(t-\tau) g(\tau)}$$

3.b Growth rate formulation

The infection incidence at time, $I(t)$, can be written in terms of the infection incidence at earlier times, and the epidemic growth rate, $r(t)$:

$$I(t) = I(t-1) \times \exp(r(t))$$

This can be rearranged to give the epidemic growth rate between two time-steps:

$$r(t) = \log(I(t)) - \log(I(t-1))$$

3.c Delayed time-series

Many epidemic time-series will exhibit delays from infection incidence to measurable outcomes (e.g. cases, hospitalisations). To quantify trends in infection incidence the delay distribution needs to be quantified and accounted for. An epidemic time-series can be written in terms of the infection incidence at previous times and the delay distribution from infection to measurable outcome, $p(\tau)$:

$$C(t) = \sum_{\tau=0}^{\infty} I(t-\tau) p(\tau)$$

where $C(t)$ is the epidemic time-series (e.g. cases). By linking the observed epidemic time-series to the time-series of infection incidence the effective reproduction number and epidemic growth rate can be quantified using statistical methods.

Suggested further reading and references

Vynnycky E, White RG. *An Introduction to Infectious Disease Modelling* (2nd ed.)
A textbook outlining a step-by-step introduction to infectious disease modelling with relevant policy examples.
https://anintroductiontoinfectiousdiseasemodelling.com/

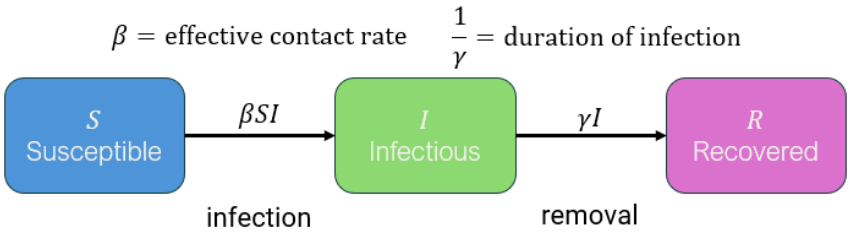
Campbell, P., Geard, N., Glass, K., McBryde, E., McCaw, J., McVernon, J., Moss, R., Ross, J., and Wood, J., 2016. A User's Guide to Infectious Disease Modelling. University of Melbourne. https://prism.edu.au/wp-content/uploads/2016/09/PRISM_Modelling-guidelines-web.pdf
A guide to help policymakers understand how to interpret the outputs of infectious disease models.

Centres for Disease Control. CDC Modelling Handbook.
https://www.cdc.gov/cfa-modeling-and-forecasting/modeling-handbook/index.html

Practical 1

This worksheet demonstrates a Susceptible-Infectious-Recovered (SIR) infectious disease model, allowing you to investigate properties of the model as disease-related parameters are changed. You will also be able to compare the deterministic and stochastic versions of the model.

Using Microsoft Excel, we can formulate the model as either a set of deterministic difference equations, where the outputs are the same every time (the deterministic model), or as a set of difference equations where we incorporate stochasticity (or randomness) into the infection and removal processes. The model structure can be drawn as a schematic diagram as in the figure below.



The spreadsheet

In the top left of spreadsheet, there are parameters defining the model output that you can adjust. These include the population size (N), the basic reproduction number (R_0), the infectious period in days (1/gamma), and the initial number of people infected.

Below this are parameters that are derived from these input parameters. These have been locked for editing but are visible so you can see how they change with the input parameters. These include the effective contact rate (beta, equal to $R_0 \times$ gamma) and the recovery rate (gamma, 1/infectious period).

On the right are two charts – the top chart shows the epidemic curve (of prevalence, not incidence) using a deterministic model, and the bottom chart shows the epidemic curve (of prevalence, not incidence) using a stochastic model. Both use the same colours: blue for susceptible people, S(t), green for infectious people, I(t), and pink for recovered people, R(t), through time.

As discussed in the lectures, the stochastic model will look different every time you run it.

You can rerun the stochastic model at any time by hitting the "F9" button, or by selecting "Formulas", then "Calculate Sheet".

Practical exercises

1. Start with a population size of 1000, a R_0 of 2.5, an infectious period of 5 days, and 10 people initially infected. This is the base scenario that we will return to throughout this sheet.

   First, look at the resulting parameters. What is the value of beta? What is the value of gamma?

   Next look at the model formulas in columns O:U, rows 5 and 6. Can you work out how the formulas relate to the schematic model diagram above?

2. Next, look at the deterministic model output. What is the final epidemic size (i.e. proportion of the population infected during the epidemic)? How long before the epidemic has died out?

   Note: Hover your cursor over the S(t), I(t), and R(t) curves to read off specific values.

3. Now look at the stochastic version of this model. Does it give similar results? How much does the final epidemic size and the timing of the epidemic change as you re-run the model by hitting F9? (Add rows to the table as necessary).

Run #	Final epidemic size	Peak timing	End of epidemic
1			
2			
3			
4			
5			
6			

4. Increase R_0 to 4 (beta = 0.8) and explore what effect this has on the final epidemic size and timing in both the deterministic and stochastic models. What has changed from the scenario where the R_0 was 2? Does the stochastic model change more/less with $R_0 = 4$?

   Deterministic model:

   Stochastic model:

Run #	Final epidemic size	Peak timing	End of epidemic
1			
2			
3			
4			
5			
6			

5. Now decrease R_0 to 1.25 and the infectious period to 2.5 days (beta = 0.5 and gamma = 0.4). What is the final epidemic size? How long does the epidemic last using the deterministic model? What about the stochastic model? Run the stochastic model a few times by pressing F9 and compare findings.

   Deterministic model:

   Stochastic model:

Run #	Final epidemic size	Peak timing	End of epidemic
1			
2			
3			
4			
5			
6			

6. We will now investigate the effect of the infectious period for a fixed value of R_0. Increase and decrease the duration of the infectious period (to keep R_0 fixed). What happens to the timing of the epidemic and the final epidemic size in the deterministic model?

R_0	Infectious period	Beta	Gamma	Peak timing	Peak size	Final epidemic size
4	5 days	0.8	0.2			
4	4 days	1.0	0.25			
4	2.5 days	1.6	0.4			
4	8 days	0.5	0.125			
4	10 days	0.4	0.1			
4	20 days	0.2	0.05			

7. Return to the scenario where R_0 = 2.5 and the infectious period = 5 days (beta = 0.5 and gamma = 0.2). Now we are going to explore the effect of the population size. Keep the initial number of people infected at 10 but decrease the population size from 1000 to 100. What happens to the deterministic model? What about the stochastic model?

8. Now we will explore the effect of the initial number of people infected. Start with our base scenario of R_0 = 2.5, infectious period = 5 days, a population size of 1000, and 10 initial infected people. Now decrease the number of initial people infected to one. What happens to the deterministic model? Run the stochastic model a few times by hitting F9. How do things change? What is happening here? Try gradually increasing the number of initial cases and see what happens.

9. Now we will investigate the relative timing and size of epidemics using parameters consistent with common human pathogens.

Pathogen	R_0	Infectious period	Peak timing	Peak size	Final epidemic size
Influenza	1.5	2.5			
COVID-19 (ancestral)	4	5			
Measles (lower bound)	16	8			
Measles (upper bound)	20	8			

10. Throughout this worksheet, we have focused on prevalence *I(t)*. Of course, for many diseases we usually (only) have knowledge of *incidence*, the number of newly recorded infections in a specified time period (perhaps also per 1,000 population). For example: the number of new cases of influenza reported over a 24hr period (the daily incidence). *Note this would also typically be reported daily, although there is no strict requirement to have the same period of time for calculating incidence and for the reporting frequency.*

    Firstly, try to write down an equation for the incidence from the SIR equations. *Note: this is not straightforward!*

    Secondly, consider how to compute this quantity in the spreadsheet. *Note: this is also rather tricky, and certainly tedious to do in Excel.*

Worksheet prepared by Freya Shearer, Alexandra Hogan, and Oliver Eales, based on original concepts by PRISM investigators.

Practical 2

Background
In the second practical, we will be introducing the programming language R, using the interface R Studio. We will then use R to implement the SIR models explored in Practical 1.

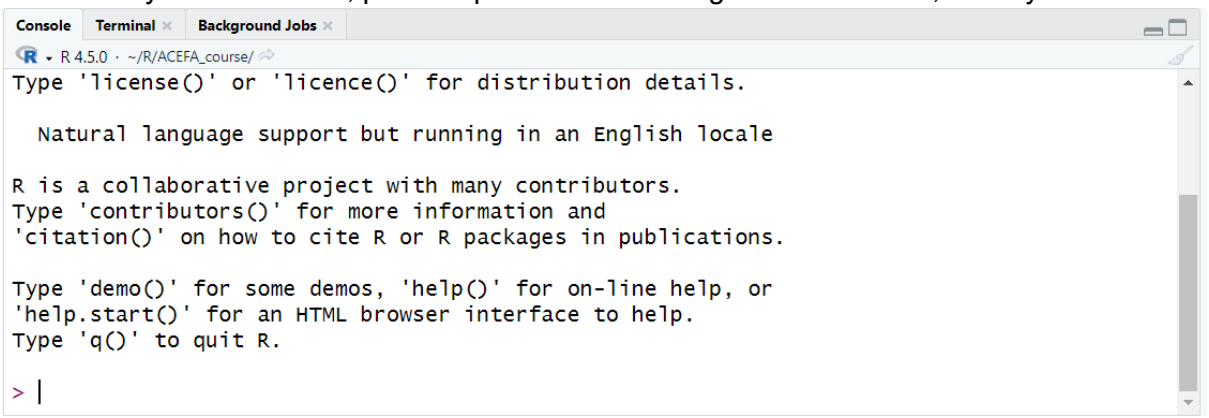
To prepare for the practical, please install the R programming language and R Studio.

To install R, follow the instructions at:
https://cran.r-project.org/bin/windows/base/

To install R Studio, follow the instructions at:
https://posit.co/download/rstudio-desktop/

Once these are installed, please open R Studio. If you have not used R before, you might like to explore the interface and try it out – there are many online guides to using R and R Studio. We will also have a section in this practical on using R, but any prior experience will be helpful.

To be ready for the course, please open R Studio and go to the console, usually on the bottom left:



To install the required packages, type in each of the following lines (hit Enter after each line):

```
install.packages("tidyverse")
install.packages("ggplot2")
install.packages("patchwork")
install.packages("EpiNow2")
install.packages("deSolve")
```

Optional practical sections
For those looking for an R refresher, you can work through the following practical sections:
part0_R_basics.R
part1_R_plotting.R

In the R basics section, you can work through an introduction to using R in R Studio. You will look at basic operations and assignments, functions, working with dataframes, and reading in data.

In the R plotting script, you can work through how to create some plots using a package called ggplot2.

The practical

In this R tutorial, we will replicate the deterministic and stochastic SIR models that we explored in the first practical. This will allow us to examine how the implementation of SIR models translates from a spreadsheet format into R code, and we will see how a number of parts of model implementation and analysis become easier in R.

To start, open the file
part2_SIR_models.R

Deterministic model

1. To start with, we are going to implement the deterministic version of the SIR model. However, instead of formulating the model as a series of difference equations (in discrete time), we are going to implement the model as a set of differential equations, in continuous time, using an ordinary differential equation solver package called "deSolve".

   Start by stepping through the code and familiarising yourself with the model framework, until you reach the part where you plot the model outputs.

2. Can you calculate the final attack rate?

3. Can you calculate the timing of the epidemic peak?

4. Can you calculate the basic reproduction number, and the effective reproduction number over time? Create a plot of R_{eff} over time.

5. Increase R_0 by setting beta = 0.8, and rerun the model. Does the model output change as expected?

Stochastic model

6. Now we are going to implement the stochastic version of the SIR model, in the same fashion as was implemented in Excel. Change the beta back to the original value of 0.4. Step through the stochastic implementation of the model and plot the model output.

7. What happens if you change the total population size? Try running the model with $N = 100$, or $N = 10{,}000$. It can be helpful to compare the deterministic and stochastic outputs by plotting on the same figure panel.

Visualisations

8. Test out the code to plot both the deterministic and stochastic model outputs on the same graph.

9. It can be useful to visualise a number of stochastic simulations on the same graph – experiment with different numbers of runs.

10. What other visualisations can you create using these model outputs?

Prevalence and incidence

11. In the last practical, we noted that we have focussed on modelling prevalence $I(t)$ and discussed how we might instead model infection incidence. See if you can work out how you might calculate incidence for the deterministic model.

Endemic infections

12. Can you modify the equations to produce endemic infection, by introducing waning immunity into the model? Hint: you will need to introduce a new parameter and change the equations. Can you also produce simulations over a longer time scale?